SAMARTH MURALIDHARA

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EDUCATION

Birla Institute of Technology and Science (BITS), Pilani (CGPA: 9.24/10.00) Pilani, India
B.E. Mechanical Engineering and Minor in Data Science Expected Jul. 2027

- **Selected Coursework**: Data Mining in Mechanical Sciences, Machine Learning, Applied Statistical Methods, Vibrations & Control, Engineering Optimization, Autonomous Mobile Robotics
- **Academic Honors**: Awarded the 1968-1973 Batch Merit Scholarship twice for academic excellence.

Vidya Mandir Senior Secondary School, Mylapore (Result: 95%) Chennai, India
CBSE Class XII May 2023

SKILLS

Programming	Python, MATLAB
Frameworks & Libraries	PyTorch, TensorFlow 2, SciPy, NumPy, ROS Noetic
Simulation & Engineering Tools	Seeq (Workbench & Data Lab), Fusion360, Gazebo, Linux (Ubuntu)
Spoken Languages	English (Fluent), Kannada (Native Speaker)

RESEARCH EXPERIENCE

Neuromorphic Computing for Structural Health Monitoring Aug. 2026 - Present
Bachelor's Thesis Research Intern, Mentor: Dr. Sayan Gupta, Professor & HoD, Applied Mechanics, IIT Madras

· Implementing Chaotic Oscillator Associative Memory architecture for spacecraft propulsion fault localization.
· Compressing multi-sensor pressure signals into phase-normalized modal features via Dynamic Mode Decomposition.
· Benchmarking COAM retrieval against conventional ML algorithms, including Random Forest and Isolation Forest.

Graph-Based Multi-Agent RL for Cooperative Navigation Jan. 2026 - Oct. 2026
Undergraduate Researcher, Mentor: Dr. Avinash Gautam, Associate Professor, BITS Pilani

· Integrated GNN-based spatial encoding & QMIX to resolve multi-agent traversal constraints in risky environments.
· Engineered a reward function balancing team traversal cost minimization with individual goal attainment.
· Reduced execution time by >99.0% compared to classical baselines (CJSG/JSG) across varying risk densities.

INTERNSHIP EXPERIENCE

Steam Turbine Performance Prediction using a Neural Network [Presentation] May 2026 - Jul. 2026
Machinery Intern, ExxonMobil, Bengaluru

· Built a 3-layer PINN to estimate the LP isentropic efficiency of a steam turbine without wet-steam dryness data.
· Trained on 8 input features across 2 years of plant data, using a custom MSE loss that uses turbine power output.
· Achieved $R^2 = 0.9906$ and RMSE = 0.95% of rated-power on unseen data and deployed a historical backfilling pipeline for a live digital twin on Seeq Workbench.

Kinematic Synthesis & Visualization of Flap Actuation Mechanism [GitHub] May 2025 - Jul. 2025
Research Intern, Mentor: Mr. Ganesh Madhuranath, Senior Principal Scientist, CSIR-NAL, Bengaluru

· Synthesized a four-bar linkage with a ternary coupler used in actuating Boeing 777 outboard flaps.
· Derived complex-valued loop closure equations for the linkage to realize takeoff, cruise & landing flap configurations.
· Validated link geometry with MATLAB & MotionGen, achieving $< 1\%$ dimensional error from Boeing 777 design data.

PROJECTS

Hierarchical Fault Diagnosis of Spacecraft Propulsion Systems [GitHub]
Course: Data Mining in Mechanical Sciences (Fall 2025), BITS Pilani

- Developed a hierarchical fault diagnosis ML framework for spacecraft propulsion with 1 kHz pressure sensor data.
- Extracted frequency-domain features via FFT to isolate water hammer effects and acoustic ringing.
- Built an RF, KNN and Isolation Forest pipeline, achieving 78.26% fault detection and 76.02% exact-match accuracy on unseen data.

Statistical Analysis and Forecasting of Solar Energy in Rajasthan, India [GitHub]
Course: Applied Statistical Methods (Fall 2025), BITS Pilani

- Conducted exploratory data analysis on 15 years of hourly solar irradiance data from two solar parks in Rajasthan.
- Implemented & benchmarked AR, MA, ARMA, ARIMA, SARIMA to predict future irradiance (GHI) values.
- SARIMA best captured geographic variations and long-term seasonality, achieving an RMSE of 19.52 W/m^2.

Reactive Robot Navigation via Vector Field Histogram & DBSCAN [Report]
Course: Autonomous Mobile Robotics (Spring 2026), BITS Pilani

- Architected a ROS Noetic & Gazebo navigation stack using odometry homing & VFH for local path planning.
- Integrated DBSCAN 2D LiDAR clustering to isolate moving threats (3–15 points) from static walls (40+ points).
- Deployed a reactive gradient forcefield for dynamic obstacle avoidance, achieving 100% collision-free navigation.

LEADERSHIP ROLES

Student Faculty Council Member - Mechanical Engg., BITS Pilani Aug. 2025 - May 2026

Orchestrated semester review meetings with faculty to address academic concerns & enhance course delivery.

Captain - Pilani Caracals Ultimate Frisbee Club, BITS Pilani Aug. 2025 - May 2026

Led a 22-member squad at College Nationals 2026, overseeing training sessions and strategic roster selection.

Treasurer - Kannada Vedike, BITS Pilani Aug. 2024 - May 2025

Managed event finances for Nalapaaka, an annual food festival attracting 600+ students from across campus.